

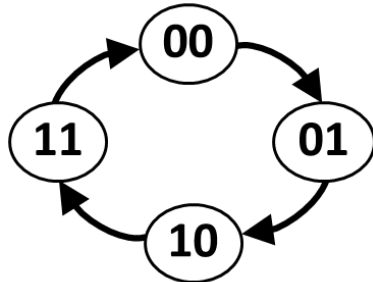
Tutorial 9 (SOLUTIONS)

NOTE THAT THERE ARE OTHER POSSIBLE SOLUTIONS TOO...

Question 1

(i) Mod-4 counter with the count sequence 0, 1, 2, 3, 0...

1) State Transition Diagram



2) Next State Table

Q_1	Q_0	Q_1^+/D_1	Q_0^+/D_0
0	0	0	1
0	1	1	0
1	0	1	1
1	1	0	0

$$D_0 = Q_0^+ = \overline{Q_1} \overline{Q_0} + Q_1 \overline{Q_0} = \overline{Q_0}$$

$$D_1 = Q_1^+ = Q_1 \oplus Q_0$$

(ii) Combinational Logic

Q_1	Q_0	C_3	C_2	C_1	C_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

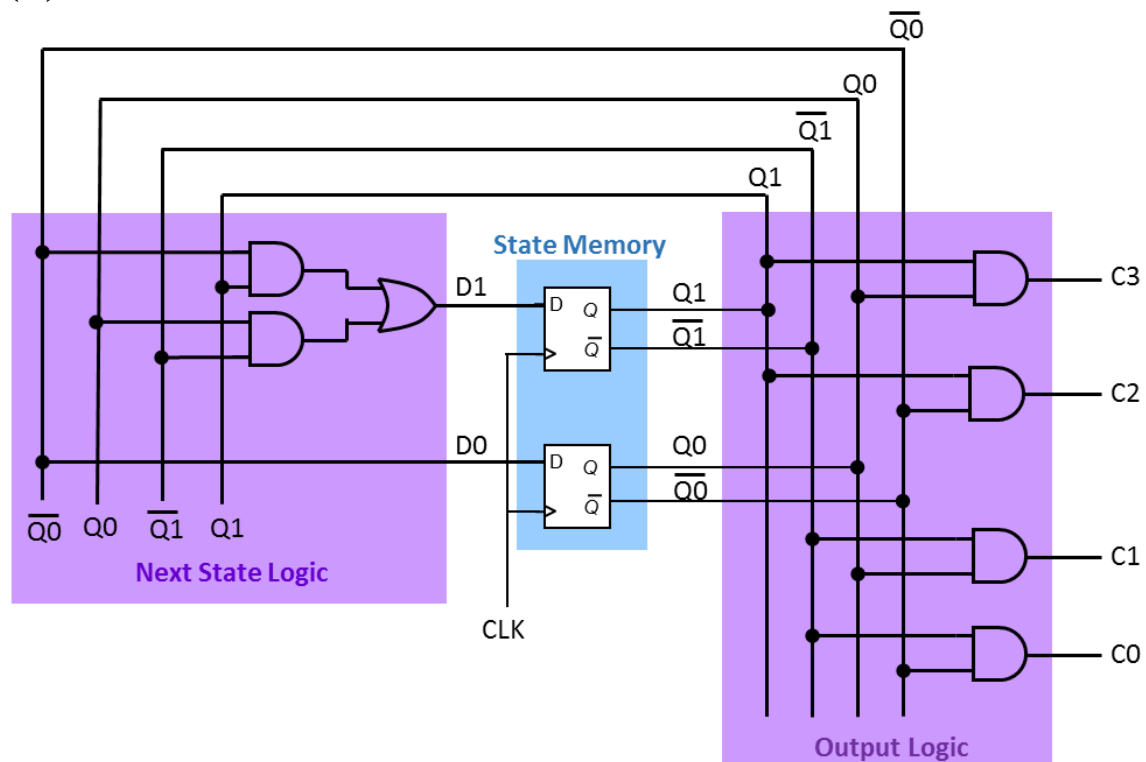
$$C_3 = Q_1 Q_0$$

$$C_2 = Q_1 \overline{Q_0}$$

$$C_1 = \overline{Q_1} Q_0$$

$$C_0 = \overline{Q_1} \overline{Q_0}$$

(iii) Circuit Realization



(iv) Comparing this implementation to T8Q2, it can be seen that while less FFs are required, more logic gates are used. Assuming the same timing characteristics for the D-FFs, the maximum operating speed of this counter would also be slower due to the need to cater for additional logic gate delays from the next state logic block.

Question 2

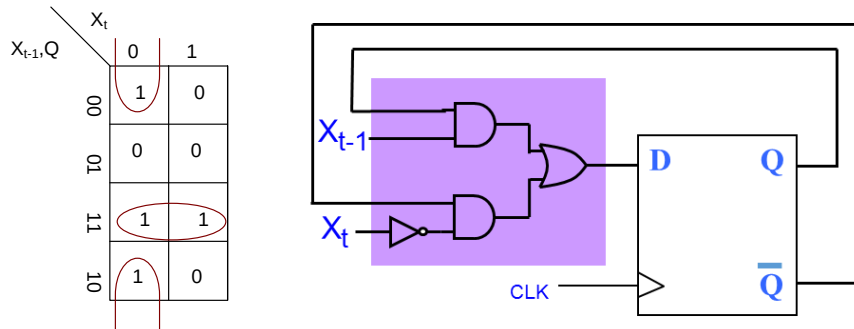
i) Design based on D-FF

The next state table (shown on the right) for X_{t-1} , X_t , Q , Q^+ is derived from the problem specification.

Since a D-FF is being used, Q^+ is also equivalent to the input of the D input of the FF. Thus, the next state table is also the truth table for the combinational circuit.

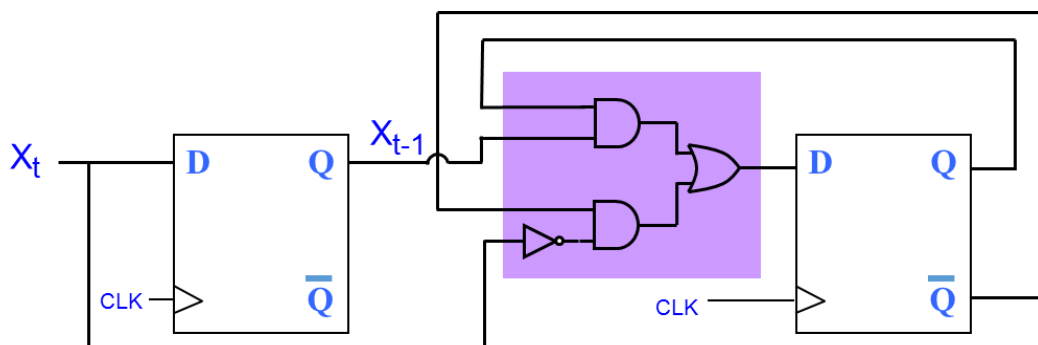
$X(t)$	$x(t-1)$	Q	Q^+ / D
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Using a D-FF, the Kmap for the combinational circuit is as below. $D = \bar{X}_t \bar{Q} + X_{t-1} Q$

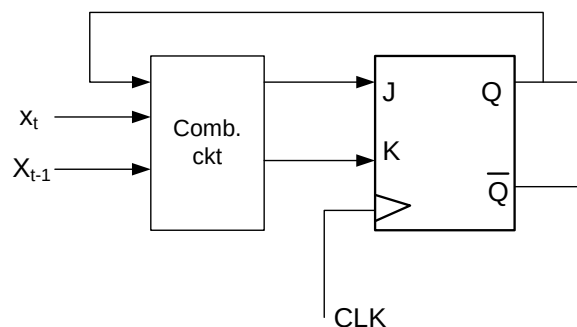


Now, how to get X_{t-1} from X_t ?

Simply connect X_t to the input of a D-FF, the output of the D-FF is naturally X_{t-1} .



(ii) Design based on J-K FF



Next state table :

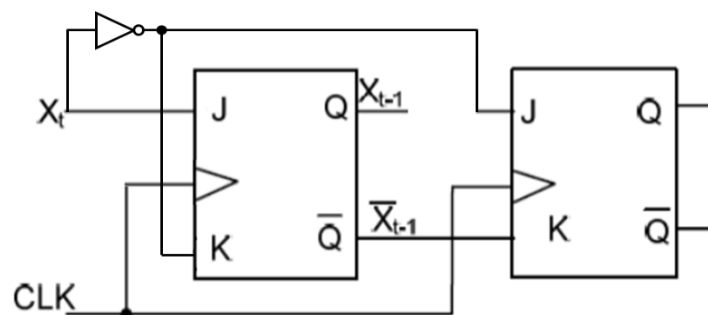
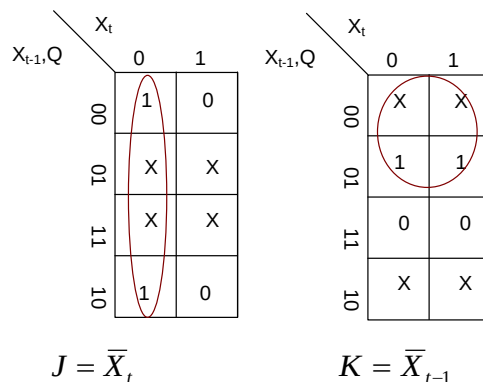
We need to apply the excitation table of the J-K FF to obtain truth table for the combinational circuit.

Q	Q ⁺	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

X(t)	x(t-1)	Q	Q ⁺
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

The truth table for the combinational circuit is:

X(t)	x(t-1)	Q	Q ⁺	J	K
0	0	0	1	1	x
0	0	1	0	x	1
0	1	0	1	1	x
0	1	1	1	x	0
1	0	0	0	0	x
1	0	1	0	x	1
1	1	0	0	0	x
1	1	1	1	x	0

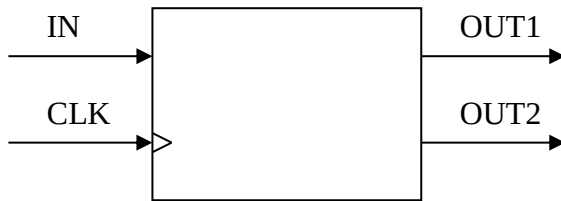


Comparison:

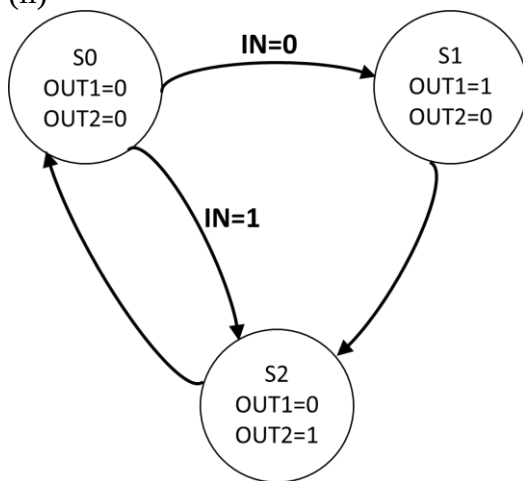
The D-FF design needs more components than J-K FF design because the D-FF is capable of only very simple behavior compared to J-K , i.e. $Q^+=D$ vs. hold, set, reset and toggle.

Question 3

(i)



(ii)



(iii)

Next State Table

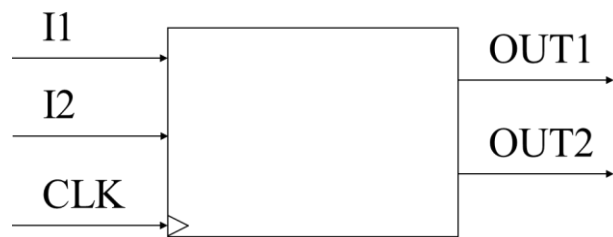
Current State	Inputs	Next State
S	IN	S+
S0	0	S1
S0	1	S2
S1	X	S2
S2	X	S0

Output Logic Table

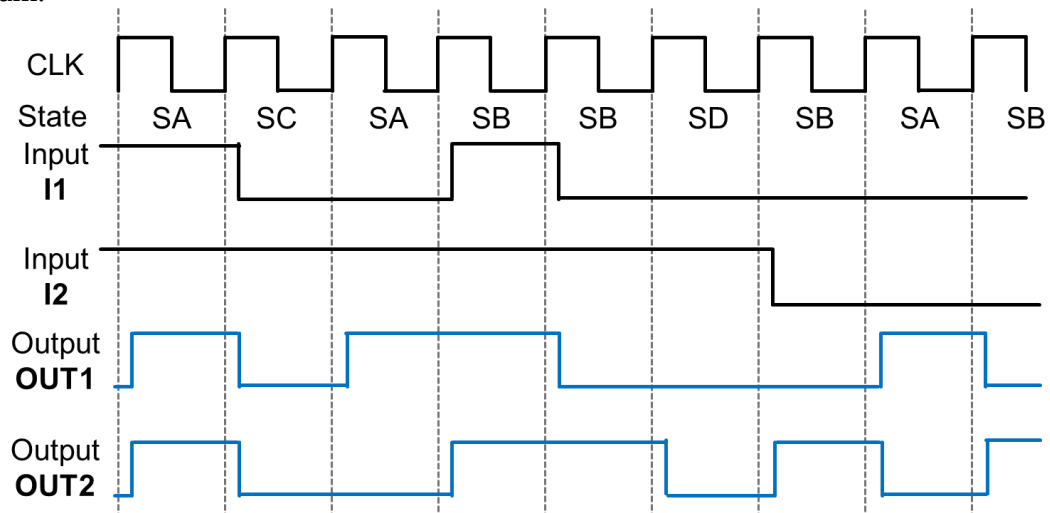
Current State	Outputs	
S	OUT1	OUT2
S0	0	0
S1	1	0
S2	0	1

Question 4

(i)



(ii) To demonstrate the effect of propagation delays, they have been included in this timing diagram.



(iii)

Next State Table

Current State	Inputs		Next State
S	I1	I2	S+
SA	0	X	SB
SA	1	X	SC
SB	0	0	SA
SB	0	1	SD
SB	1	X	SB
SC	X	X	SA
SD	X	X	SB

Output Logic Table

Current State	Outputs	
S	OUT1	OUT2
SA	1	I1
SB	I1	1
SC	0	0
SD	0	0